

TA brief

No cold calling; nobody is asked to read difficult text aloud.

Beside the You do, mark I (independent), S (supported) or C (choice): the level of independence, not the effort.

Give time to start → point to the relevant place → rephrase → offer a small prompt or word bank → reduce to a limited choice. Never straight to telling the answer.

Tell the teacher the barrier: reading, vocabulary, processing, regulation, writing or a science misconception. Avoid prompting through every question or dictating the final answer.

If behind

A pupil who missed last lesson: give the Arrival reminders and today's must-know words, not the missed lesson's work.

Photograph the practical at the planned point, upload it to the school's evidence app with this code, and stick a copy in the book.

Lesson code: GROW-SCI-A1-W08-P2. Planned photograph: after recording observations and before the conclusion.

P17 · SLT planning and evaluation

Lesson code : GROW-SCI-A1-W08-P2

SoW reference : GROW scheme of work, Autumn 1, return lesson

Exact SoW outcome : Source lesson outcomes: Explore friction and how it can be helpful or unhelpful.

Investigate mechanisms: levers, pulleys and gears. Plan and carry out a fair test about forces; record results. Describe the movement of the Earth and other planets around the Sun. Describe the movement of the Moon around the Earth.

Derived review purpose : Review learning from forces, mechanisms, fair enquiry, Earth and the Moon. This review purpose is assembled from the five source lesson outcomes.

Evidence (scheme of work) : Pearson Edexcel Entry Level Certificate in Science 8939 (E1 → E3); AQA UAS science units. Working-scientifically observation, explanation and recorded evidence.

Evidence in this lesson : Explain one helpful and one unhelpful friction example.; Explain a mechanism gain and its cost.; Choose change, measure and controls for a fair comparison.; Explain an orbit and the Moon's reflected light.

Evidence location : Book work beneath the strip, retained first answer and pupil-owned edit. Keep spoken, practical or photographic evidence under the same code.

Label evidence on the school's evidence app with this code : GROW-SCI-A1-W08-P2

Arrival task (4 min) : Task: In your book: Glue in your strip; answer four prompts. Staff move: Read four prompts, sample privately and choose one route. IF NEW: use provided reminders. IF REVISION: retrieve before opening help.

Starter · choose a first idea (4 min) : Task: In your book: Keep one first idea and a reason. Staff move: Receive a first idea without upgrading the pupil's meaning; return to it after the evidence.

I do · make the science visible (4 min) : Task: In your book: Watch the model; name the evidence. No writing yet. Staff move: Keep the visible model and language linked. Pause at one observation; distinguish what the model shows from its limit.

We do · everyone answers (4 min) : Task: In your book: Choose an answer; explain the evidence clue. Staff move: Everyone commits. Secure: transfer. Mixed: compare two choices. Misconception: re-model, then use a fresh example. Access: change response mode.

I do · connect the evidence (3 min) : Task: In your book: Watch the connection; name the limit. No writing yet. Staff move: Model the evidence-to-reason connection; keep the stated limitation visible.

We do · Connect each claim to its science (5 min) : Task: In your book: Record one placement and its science reason. Staff move: Pupil operates or directs. Name back the clue before any correction. Use the lightest prompt and fade it.

You do · explain and improve (9 min) : Task: In your book: Record evidence, explain it, then improve one part. Staff move: One adult supports the pupils needing access; the other holds the group, then swap roles across the week. Keep the first answer; give VF and one NS+; pupil edits // in the same colour on the same page.

Review and improve (3 min) : Task: In your book: Review one explanation; make one edit and mark it //. Staff move: Use three minutes: hear one precise VF/NS+ (one minute), make a pupil-owned edit (one minute), then receive the explanation of the change (one minute). Keep the original meaning, use // and add R only after genuine adult audience.

Exit ticket (4 min) : Task: In your book: Answer the prompt; show one next step. Staff move: Read the response privately. Pupil voice plus adult audience closes R only when both happen. Use the response to choose the next teaching route.

Practical safety : Use prepared diagrams and cards. No new practical is required; handle any models under usual classroom routines.

Preparation and equipment : Genuine books, prepared evidence cards and familiar models.

Planned adaptation — Cognition and learning : Arrival and independent: one decision at a time; show a worked example, provide the topic word bank and fade the scaffold when the reason is secure.

Planned adaptation — Communication and interaction : We do: private thinking and partner rehearsal; accept pointing, signing, drawing, speaking or exact scribing. The pupil directs the decision.

Planned adaptation — SEMH : Predictable route choice, quiet pause and small re-entry step. Receive uncertainty calmly and keep correction private.

Planned adaptation — Sensory and physical : Model and practical: large diagrams, stable equipment, seated or no-touch director route; follow the topic safety note.

Planned adaptation — EAL : Pre-teach the named science words with the model; permit first-language planning and rehearsal. EAL is not SEN and does not reduce the science goal.

EEF strategy line : Explicit instruction in I do; scaffolding in the independent frame; flexible grouping for rehearsal; metacognitive checking in review; technology or diagrams to remove access barriers.

Staff: confirm the planned adaptations actually used : Record which adaptation helped and why in the evaluation.

If behind — Must teach core : I can explain familiar science using an observation, model or genuine piece of work.

If behind — Shorten or merge : Shorten repeated rehearsal after a secure hinge; retain the individual explanation, safety and exit.

If behind — Next lesson : Use this evidence to choose the next topic route.

If behind — Dependency : Pass on the checked explanation and unresolved misconception; a group product cannot replace individual evidence.

Staff evaluation — Date taught : Record the date taught.

Staff evaluation — Delivered as planned : Yes / Partly / No; explain the change.

Staff evaluation — Criteria met : All / Most / Some; name the criterion evidenced.

Staff evaluation — Adaptations made and why : Record the support used and its effect.

Staff evaluation — Change from the SoW and coverage impact : Name the content affected and the next teaching action.

Staff evaluation — Evidence banked : Record the book location or evidence labelled on the school's evidence app.

P18 · Individual target opportunities

Cognition and learning : Arrival and independent: one decision at a time; show a worked example, provide the topic word bank and fade the scaffold when the reason is secure. Use the pupil's own target and note the prompt level, not a diagnosis.

Communication and interaction : We do: private thinking and partner rehearsal; accept pointing, signing, drawing, speaking or exact scribing. The pupil directs the decision. Use the pupil's own target and note the prompt level, not a diagnosis.

SEMH : Predictable route choice, quiet pause and small re-entry step. Receive uncertainty calmly and keep correction private. Use the pupil's own target and note the prompt level, not a diagnosis.

Sensory and physical : Model and practical: large diagrams, stable equipment, seated or no-touch director route; follow the topic safety note. Use the pupil's own target and note the prompt level, not a diagnosis.

EAL : Pre-teach the named science words with the model; permit first-language planning and rehearsal. EAL is not SEN and does not reduce the science goal. Use the pupil's own target and note the prompt level, not a diagnosis.

Individual plans : Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan. Follow the pupil's IBSP where it differs from these routines.

First lesson back · staff card

- Before the lesson, scan the books for five minutes. Look for genuine coverage and unfinished explanations; do not assume a missing page means the pupil was absent.
- Ask the cover teacher and TAs what was taught, what was practised and what still needs teaching.
- Check the roll against the books so that absence and missing evidence are kept separate.
- Existing NS+ is history. Begin fresh feedback from what the pupil shows now; do not retrospectively close an old loop.
- After the check, use one read, once, to shape Autumn 2. Choose the next teaching route from the pupil's evidence; do not add another record.

Written RAG rule · each success criterion

Explain one helpful and one unhelpful friction example. : Green — accurate independent explanation with relevant evidence; Amber — partial explanation or correct with a prompt; Red — missing explanation, misconception or access still unresolved. Green → transfer to a new example. Amber → one modelled example then a fresh attempt. Red → clarify access, re-teach this idea and check privately.

Explain a mechanism gain and its cost. : Green — accurate independent explanation with relevant evidence; Amber — partial explanation or correct with a prompt; Red — missing explanation, misconception or access still unresolved. Green → transfer to a new example. Amber → one modelled example then a fresh attempt. Red → clarify access, re-teach this idea and check privately.

Choose change, measure and controls for a fair comparison. : Green — accurate independent explanation with relevant evidence; Amber — partial explanation or correct with a prompt; Red — missing explanation, misconception or access still unresolved. Green → transfer to a new example. Amber → one modelled example then a fresh attempt. Red → clarify access, re-teach this idea and check privately.

Explain an orbit and the Moon's reflected light. : Green — accurate independent explanation with relevant evidence; Amber — partial explanation or correct with a prompt; Red — missing explanation, misconception or access still unresolved. Green → transfer to a new example. Amber → one modelled example then a fresh attempt. Red → clarify access, re-teach this idea and check privately.

The receiving teacher uses each criterion to set the next routes. No overall colour replaces the separate science decisions. Missing evidence is not evidence of completion.

Preparation and practical safety

Genuine books, prepared evidence cards and familiar models.

Use prepared diagrams and cards. No new practical is required; handle any models under usual classroom routines.

Use prepared diagrams and cards. No new practical is required; handle any models under usual classroom routines.

Staff cut the activity pieces before the lesson. Move pieces for the task; pupils record the answer and reason in their books.

Illustrative data are reasoning examples only, never a record that a pupil completed the practical. Use actual observations for completion evidence.

Adult deployment and feedback

During the independent task, one adult works with supported pupils and the other holds the group; swap the role across the week. Use least-prompt-first. Pupil meaning stays separate from adult support.

VF: verbal feedback. WS: worked with support. I: independent. NS+: one specific next step. E: evidence on the school's evidence app. //: pupil self-edit. ?: read this back. R: pupil response and genuine adult audience both happened.

Voice + Audience = R. The pupil owns the response; the adult reads or hears it and names the change. No extra receipt, signature or retrospective closure. Keep the edit in the same colour on the same page.

Hinge routes

Secure: transfer. Mixed: compare two choices. Misconception: model a counterexample, then ask a fresh question. Access: change the response mode. Method or data concern: restore the control or locator before judging the claim.

Answers and teaching cues

Hinge

Keep block, release point, ramp and measurement method the same. Change the surface and measure distance in centimetres.

Model steps and limits

- Use genuine work if it is available; a missing record stays missing.
- A supplied comparison says the same block travelled 45 cm on smooth board and 19 cm on fabric. These are illustrative data, not your practical result.
- The evidence can support a claim about these surfaces and this block; one example is not every surface.
- A lever may reduce the effort force by increasing the effort distance. It does not create energy.
- Gravity pulls planets inward and bends their paths. Earth is third from the Sun.
- The Moon reflects sunlight. Changing view of its lit half creates phases; a lunar eclipse involves Earth's shadow.
- Illustrative data support reasoning practice only. They are never recorded as a completed practical.

Activity answers

- Brakes: Friction helps the brakes slow movement.
- Lever: A mechanism trades distance for force.
- Controls: One planned variable changes.
- Moonlight: The Moon does not make its own visible light.
- Orbit: Gravity bends the moving planet's path.
- Engine: Friction can hinder a machine while helping brakes.

Misconception repair

A scientific claim needs relevant evidence, a reason and a fair limit.

Arrival answer cues

Supported

- 1. Friction helps shoes grip.
- 2. The pivot is the point around which the lever turns.
- 3. Keep the block the same while changing the surface.
- 4. The Moon reflects sunlight.

Independent

- 1. Brakes/shoes: helpful resistance; engine rubbing: unwanted resistance and heating.
- 2. The fixed point the lever turns around.
- 3. Same block, ramp, release point and measurement method.
- 4. Gravity curves Earth's path; reflected sunlight lights the Moon.

Aspire

- 1. For example, brake pad and disc touch; friction opposes their relative sliding.
- 2. Less effort force requires more effort distance for the same load rise.
- 3. Two planned changes prevent isolating which caused the difference.
- 4. Normal phases reflect viewing geometry; Earth's shadow causes a lunar eclipse.

Exit and independent answers

Check the pupil's own evidence against the learning criteria. Explain one helpful and one unhelpful friction example.; Explain a mechanism gain and its cost.; Choose change, measure and controls for a fair comparison.; Explain an orbit and the Moon's reflected light.

Retain first and revised versions. A claim must name relevant evidence and a scientific reason; keep the stated model limit. Supplied examples are not personal practical completion.

Further topic teaching prompts by stage

Title

- Friction: Friend and Enemy
- last lesson — Explore friction and how it can be helpful or unhelpful.

- I can state that friction opposes sliding, or attempted sliding, between touching surfaces.
- I can give one example where friction is helpful and one where it is unhelpful.
- I can predict which surface will produce more friction, and say why.
- Lesson A resources Lesson B resources
- Make participation possible
- Use regulation time, a trusted adult, accessible response routes and permission to refuse. The small help prompt is optional; useful feedback remains part of ordinary teaching. Refusal is information, not a reason to invent a response or punish passing.
- Lesson A: a claim worth revisiting
- During We Do 2, invite a pupil to identify one word or idea in a sorted claim that needs checking. Hear, read or watch the response. Use the actual surface-pair example to adapt the explanation, scaffold or next question. Keep this within the existing eight minutes of sorting and correction, then use two minutes for the exit.
- Lesson B: review the evidence
- After the 32-minute workshop, use four minutes to revisit one conclusion and its supporting results with the adult, then four minutes for the exit. Use the existing record or an equivalent response; do not require a separate loop sheet. Explain any limit on a requested change and check whether the response helped.
- Space makes participation possible; Voice is the pupil's response in their own modality; Audience requires the adult to engage with it; Influence is the resulting adjustment to explanation, task, scaffold or subsequent teaching. One optional displayed prompt does not limit genuine feedback exchanges.
- Use marking codes honestly
- Use R only after a pupil response and genuine adult engagement. A click, download, opened explanation or completed exit does not automatically establish Audience, Influence, R or an evidence check outcome. Record only the evidence and help actually provided; do not require a duplicate written record for every task. Use agreed policy codes appropriately. Extended writing or evidence check may need one or two precise next steps and response time.
- Print staff feedback guidance
- Optional existing staff records; use only when appropriate.
- Print optional marking notes
- Two separate 40-minute lessons: Lesson A: Friction: Friend and Enemy. Lesson B: Friction: test the surfaces. Keep this deck and your work for both lessons.
- Resume Lesson B — Friction: test the surfaces

Arrival Task

- Arrival task — pick your tier
- Choose one route. Point, say, sign, draw or write. You may pass or say not sure. Use screen or paper; you do not need to do both.
- For question 3, hand rubbing is optional. If comfortable, gently rub your hands for up to five seconds; stop at discomfort. You can watch an adult or describe a familiar hand-rubbing example instead. Say whether you observed it, remembered it or predicted it.
- 1. A push moves a toy car away. Which action moved it? Push / pull.
- 2. Look at two palms pressed together. Which surfaces touch? Palm and palm / palm and sleeve.
- 3. Use your chosen route. What do you notice when hands rub? Say, point or draw one observation.
- 4. A shoe needs to grip the ground. Does grip help walking or make slipping easier? Help walking / make slipping easier.
- Allow four minutes within the existing arrival stage. Read or scribe as needed. Start from what pupils show; the word friction is not required before it is taught. Accept uncertainty and equivalent responses. Discuss one useful observation rather than completing a second record. A remembered example or prediction is not a measurement made today.
- Push. Accept pointing to the word or showing the action.
- Palm and palm. Accept indicating the two touching surfaces.
- Accept what was actually noticed, including no noticeable warmth. A remembered or predicted change must be identified as such; do not invent an observation.
- Help walking. Grip helps stop the shoe slipping against the ground.
- Print Supported staff answers
- 1. Name the two actions that can change how an object moves: p____ and p____.
- 2. When hands rub, name the two touching surfaces.

- 3. Use your chosen route. Describe one change you notice or already know. If you know the force's name, add it.
- 4. A bike brake pad presses on the wheel rim. Predict what happens to the turning wheel.
- Push and pull. These are familiar ways of applying forces.
- The skin of one palm and the skin of the other palm.
- Accept an actual observation or a clearly identified remembered example. Hands may feel warmer. Friction is the optional force name; no noticeable warmth is also a valid observation.
- The wheel slows. The pad and rim resist sliding against each other. Treat the response as a prediction unless an actual demonstration is observed.
- Print Independent staff answers
- 1. A toy car is pushed, then slows on a level floor. Name one force you think could slow it. Not sure is useful.
- 2. A hand grips a door handle. Name the touching surfaces and explain what the grip helps you do.
- 3. Use your chosen route. Suggest why rubbing hands may feel warmer. Give a prediction if you cannot observe it.
- 4. Two toy cars start at different speeds on different surfaces. Is that a fair surface comparison? Name one thing to keep the same.
- Accept a plausible resistive force, such as friction in wheel or axle contacts or air resistance. Do not require a single location or teach that friction always opposes the whole object's travel.
- Hand and handle. Grip helps turn the handle without the hand slipping over it.
- Work done while rubbing transfers energy to thermal stores, so the skin may warm. Accept an initial idea to revisit; do not claim that every brief rub must produce noticeable warming.
- No. Starting speeds differ as well as surfaces. Compare using the same car and a consistent release, changing the surface only; accept one sensible control.
- Print Aspire staff answers

Today at a Glance

- Predict and test two surfaces

I Do 1

- Watch me test two surfaces
- Change one thing — the surface. In this illustration, carpet gives a shorter run and ice gives a longer run. Use the actual surface pair and evidence; texture alone is not enough. Sliding friction opposes sliding between the touching surfaces.
- friction — A force between two surfaces that are rubbing or trying to slide.
- surface — The outside face of an object, where the touching happens.
- grip — Useful friction — stops slipping.
- wear — Damage caused over time by friction.

I Do 2

- Defined friction as a force between rubbing surfaces.
- Gave one helpful and one unhelpful example.
- Predicted, tested and recorded a surface comparison.
- Capture: Photograph the pupil's surface-test results table with their prediction still visible.

We Do 2

- During the claim sort, point, say, sign or show the adult an idea, question or change that would help. You may pass or speak privately. The adult listens, explains what can change and checks whether it helped.
- Two small checks to finish Lesson A. Point, say, sign, draw, type or use paper. An adult can scribe your words. You may pass; complete one route, once.
- Left. Friction on this sliding box acts along the contact, against its slide to the right.
- Yes. Pad–rim friction helps slow the wheel. The touching surfaces and the intended job make this a helpful example.
- Use these with an adult. Your response is not automatically marked.
- Print Lesson A staff exit answers
- Surfaces. Accept a clear equivalent that identifies the two contacting surfaces.
- Slow down or stop. The brake pad and rim resist sliding against one another.

- The pad and rim contact; friction resists their relative sliding and helps slow the wheel.
- Unwanted rubbing can heat and wear moving parts. Judge friction against the intended job, not as an always-bad force.
- Next lesson: Friction: test the surfaces

Independent Work

- Your turn — pick your tier
- Direct or observe the three-surface test with an adult. Point to the start and stop marks; enter or dictate results in the supplied table and choose a supported conclusion.
- Test three surfaces with repeated releases, record distances in centimetres, and use the results to identify which surface resisted sliding most in this setup.
- Compare repeated results and an uncertain run. Explain a control or measurement improvement, retry if possible, and qualify your conclusion to these surfaces and this block.
- A surface test with evidence
- 4 min: Retrieve your Lesson A prediction. Check the fixed low ramp, the same sliding block, release mark and safe run-out; choose tester, observer or recorder.
- 20 min: Release the block without a push onto each of three level test surfaces. Measure from the same ramp-foot mark to where it stops, in centimetres. Aim for two trials per surface; record a non-slide rather than inventing a distance.
- 2 min: Pens down if you are writing; pause and check. Compare repeats and check one control variable. A different push, slope or measuring start makes a comparison uncertain.
- 6 min: Repeat one doubtful run if time allows, draw an evidence-based conclusion, and return the equipment. If trials are unfinished, record that honestly.
- Access and equipment: Adult-secured low ramp, one small sliding block, three level surface samples, ruler/tape and a tray or padded stop. Keep release height and the ramp-to-surface join fixed. Keep hands out of the run; no oil, loose powders or pupils sliding on floors. Observation/directing is an equal route.
- If equipment is unavailable: analyse this invented teaching data for the same controlled setup, labelled as model data: card 42 and 44 cm; fabric 25 and 27 cm; rubber mat 15 and 18 cm. These are not pupil measurements and do not prove a universal order for materials.
- My conclusion: _____ because _____.
- A result or control to check: _____.

Review the evidence

- Use your existing results or the clearly labelled model data. Point, speak, sign, draw or use your current work; no extra feedback form is needed.
- Choose one conclusion. Show the adult which results support it. Name the surfaces and whether the evidence is measured or model data.
- Consider the feedback. Does the conclusion fit both trials? Check one control or uncertainty; a surprising result is not automatically wrong.
- Keep or improve your explanation and show why. The adult responds to your idea or question. Keep the original results; label any later retry separately.
- You may ask for more thinking time, use a trusted adult or pass. Finish with the existing exit question; this review stays within its four minutes.
- While reviewing your evidence, point, say, sign or show the adult an idea, question or change that would help. You may pass or speak privately. The adult listens, explains what can change and checks whether it helped.

Exit Ticket

- Name one place friction helps.
- brakes / shoes / tyres / handles
- Explain why a car needs friction to stop.
- The brake pads rub the disc; friction converts movement energy to heat and slows the wheel.
- Explain why the same force can be a design problem and a design solution.
- Friction opposes relative sliding between the touching surfaces. That resistance is wanted at the brake and can be unwanted in the engine.

Title

- last lesson — Investigate mechanisms: levers, pulleys and gears.

- I can name the three mechanisms and point to one example of each.
- I can identify the pivot on a lever.
- I can explain what a mechanism gives you and what it costs you.

Arrival Task

- Open the scissors on your desk. Point to the part that everything turns around.
- Name the part of the scissors that everything turns around.
- Name the pivot on the scissors, and on the wheelbarrow picture.

Today at a Glance

- Move it and feel the difference

I Do 1

- Slide the ▲ PIVOT toward the load and the lift gets EASIER — but the hand travels FURTHER. A mechanism trades distance for force; it never gives force for free.
- mechanism — A part that changes a force — its size, its direction, or both.
- lever — A rigid bar that turns around a pivot.
- pivot — The fixed point a lever turns around.
- pulley — A wheel with a rope over it; changes the direction of a pull.

I Do 2

- Named lever, pulley and gear with one example each.
- Identified the pivot correctly.
- Stated both the gain and the cost of a mechanism.
- Capture: Photograph the pupil's lever set-up at both pivot positions, with the load visible.

Independent Work

- Direct or observe both marked pivot positions. Tick which feels easier, then point to where the effort, load and pivot are. Tell the adult if the effort felt similar.
- Compare both positions with the same load and lift height. Record easier/similar/harder and explain how the effort arm changed.
- Compare effort and hand travel for the same load rise. Measure hand travel if suitable equipment is available; use force numbers only if measured with an appropriate meter. Explain the trade-off.
- Two lever positions, one load
- 4 min: Recover the last lesson's lever rule. Check the adult-prepared lever, light load and two marked pivot positions.
- 20 min: Use the same load and effort point. Compare a middle pivot with a pivot nearer the load. Raise the load to the same small target height; record the effort as easier/similar/harder, and compare hand travel.
- 2 min: Pens down if you are writing; pause and check. Check whether the load, lift height and effort point were kept the same. A partner or adult may move the lever while you observe.
- 6 min: Repeat one comparison, improve a label or explanation, and return all small parts. Use a diagram explanation if safe equipment is unavailable.
- Access and equipment: Adult-secured card/ruler lever, broad stable pivot and a light secured load such as an eraser. Use a small controlled lift, never launch objects; hands stay clear of the pivot. Preassembled or adult-operated kit is available.
- Label on a simple sketch: EFFORT — PIVOT — LOAD.
- My explanation: _____.
- Model limit: A ruler lever shows the force–distance trade-off; friction and bending make a real model imperfect.

Lundy Loop

- Where what you say changes something
- Loads and masses get dropped. Feet clear of the bench, and tell me if the weight is too much to handle safely.
- Point, sign, say or show one observation, question or access need. You may pass or speak privately.
- Tell the adult teaching this lesson, or ask a chosen partner to help you pass on the message.
- The adult says what can change now, makes one agreed change if possible, and explains any limit. Check whether the change helped; no display or sharing is assumed.

Exit Ticket

- Name the point a lever turns around.
- Explain why moving the pivot closer to the load makes lifting easier.
- The effort arm becomes longer relative to the load arm, so less force is needed.
- Explain why no mechanism can give you force without a cost.
- A mechanism cannot create energy. Gaining output force involves a distance trade-off; real mechanisms also dissipate some energy, for example through friction.

Title

- last lesson — Plan and carry out a fair test about forces; record results.
- I can name the one thing I will change and the things I will keep the same.
- I can record results in a table with units.
- I can say whether my results answer my question.

Arrival Task

- Read the question on the board. Circle the ONE thing we will change.
- Write the one thing we will change in today's test.
- Write the one thing we will change and two things we must keep the same.

I Do 1

- Watch me spoil a test on purpose
- Plan a fair test: ◆ I CHANGE one thing, ■ I KEEP THE SAME the rest, ► I MEASURE with units — then three repeats and a mean. Icon + word on every box.
- variable — Anything in a test that could change.
- independent variable — The ONE thing you deliberately change.
- control variable — Everything you deliberately keep the same.
- dependent variable — The thing you measure.

I Do 2

- Identified independent, dependent and control variables.
- Recorded results in a table with correct units.
- Judged whether the data answered the question.
- Capture: Photograph the completed planning frame AND the results table together — the pair is the evidence.

Independent Work

- Choose or point to CHANGE / KEEP / MEASURE labels with an adult. Direct three trials on one surface, then compare with the adult-recorded second surface. Dictate results and one conclusion.
- Write or dictate your variable plan, run three trials per surface, and complete the results with centimetres. Use repeated evidence to answer the question.
- Plan and run the repeats, calculate means if ready, and explain an unusual result or limitation. A mean does not repair an unfair method; improve the method before claiming more.
- Plan, repeat, check and conclude
- 7 min: Return to the last lesson's test question. Use the first seven minutes to choose two surfaces, complete CHANGE / KEEP / MEASURE, and check setup with an adult.
- 17 min: Use seventeen minutes for three releases per surface, if practicable. Keep the trolley, ramp height and release point the same; release without a push. Record stopping distance in centimetres from the same zero.
- 2 min: Pens down if you are writing; pause and check. Use two minutes to compare repeats. Mark an unusual run and a possible method issue; do not delete a result just because it differs.
- 6 min: Use six minutes to retry one method issue if time permits, state what the evidence supports, and reset equipment. Keep both original and repeated values.
- Access and equipment: Adult-secured low ramp, same trolley, two level run-out surface samples, measuring tape and padded stop. Agree roles, a clear release signal and hands out of the run. Record or direct instead of handling if preferred.
- Result kept for checking: _____. Method issue or uncertainty: _____.
- My evidence supports _____ because _____.

- No-equipment route: an adult can provide the labelled invented teaching set A = 48, 50, 49 cm; B = 27, 29, 28 cm. Evaluate what would need controlling to trust this comparison; record this as data interpretation, not an experiment you performed.

Lundy Loop

- This lesson has a lot of waiting between trials. Waiting is part of it, not wasted time.

Exit Ticket

- How many things do you change in a fair test?
- Name one control variable from today's test.
- ramp height / trolley / release point / push force
- Explain why three repeats are better than one.
- Repeats show whether a result is consistent or a one-off, so you can judge reliability.

Title

- last lesson — Describe the movement of the Earth and other planets around the Sun.
- I can name the eight planets in order from the Sun.
- I can state that the planets orbit the Sun, and that the orbits are nearly circular.
- I can explain what keeps a planet in orbit.

Arrival Task

- Put the four planet cards in order from the Sun.
- Write the first four planets in order from the Sun.
- Write all eight planets in order from the Sun.

I Do 1

- Order from the Sun, read by icon + word: ● rocky inner planets, ○ gas and ice giants outer. The inset shows why they orbit — gravity pulls in, motion goes sideways.
- orbit — The curved path one object takes around another.
- Solar System — The Sun and everything that orbits it.
- gravity — The force pulling objects towards each other.
- year — The time a planet takes to complete one orbit of the Sun.

I Do 2

- Named the eight planets in order.
- Classified planets as rocky or giant.
- Explained gravity's role in maintaining an orbit.
- Capture: Photograph the pupil's ordered planet strip with their orbit annotation.

Independent Work

- Order the eight names with support. Direct the Earth counter around the Sun and point the gravity arrow towards it. Compare Earth and Mars using the supplied days; show which year is longer.
- Order and group the planets. Draw or build the two-arrow orbit model, use two pairs of year values, and explain why a planet keeps changing direction.
- Use the year data to test a farther-out/longer-year prediction. Explain why hand-moving counters is not an experiment on gravity, and identify a scale or speed limitation.
- Build and challenge an orbit model
- 4 min: Retrieve the last lesson's planet order with the eight names. Choose counters on a desk, drawing or directing an adult; no public walking is required.
- 20 min: Build a Sun–Earth counter model. Show the direction of motion and the direction of gravity separately. Compare the supplied year data, predict which of two planets completes an orbit first, then check against the numbers.
- 2 min: Pens down if you are writing; pause and check. Explain or point to one arrow and one data comparison for an adult or chosen partner. Check that a longer year means a longer time for one orbit.
- 6 min: Correct one arrow or comparison. Record two things the model shows and one thing it does not show; return counters and labels.
- Access and equipment: Sun/Earth labels, counters, paper arrows and the provided numerical table. Keep everything on the desk if preferred. Do not swing objects on strings. The drawing is schematic, with no accurate size, distance or speed scale.

- Original classroom diagrams and tasks; the sources below support the facts. No external page is needed to do the activity.
- NASA Space Place: years on other planets — <https://spaceplace.nasa.gov/years-on-other-planets/en/>
- NASA: about the planets — <https://science.nasa.gov/solar-system/planets/>

Lundy Loop

- Modelling an orbit means walking a circle in front of others. You can model it with counters on the desk instead.

Exit Ticket

- Which planet is third from the Sun?
- What force keeps a planet in orbit?
- Explain why a planet does not fall into the Sun.
- It is moving sideways fast enough that the inward pull curves its path instead of pulling it in.

Title

- last lesson — Describe the movement of the Moon around the Earth.
- I can state that the Moon orbits the Earth, roughly once a month.
- I can explain that the Moon does not make its own light.
- I can describe why the Moon appears to change shape.

Arrival Task

- Point to the full moon on the phase strip.
- Write how long you think the Moon takes to go round the Earth.
- Write how long the Moon takes to orbit Earth, and where its light comes from.

Today at a Glance

- Where the light comes from

I Do 1

- Watch me make the phases happen
- Torch-and-ball model: sunlight lights one half of the Moon always; as it orbits, our angle changes, so we see NEW, HALF, FULL. The phases are not the Earth's shadow.
- satellite — An object that orbits a planet. The Moon is Earth's natural satellite.
- phase — The shape of the lit part of the Moon that we can see.
- reflect — To bounce light off a surface.
- crescent — A thin curved phase.

I Do 2

- Stated that the Moon orbits Earth in about a month.
- Explained that moonlight is reflected sunlight.
- Modelled or described the cause of the phases.
- Capture: Photograph the pupil's completed phase wheel with the sunlight direction marked.

Independent Work

- Arrange or draw new, half and full Moon examples with an adult. Direct one model position and point to the lit part you can see. Correct a matching label if needed.
- Build and label a phase wheel or three-view strip. Check three positions with the model and explain why the visible lit part changes although the Moon stays round.
- Check the three views and explain the phases. Distinguish a lunar eclipse from the regular phase cycle, and explain one limitation of the torch/ball model.
- 4 min: Retrieve the last lesson's lit-side rule. Choose a pre-cut phase wheel, simple drawings or directing an adult's model; agree before any light level changes.
- 20 min: Build or arrange new, half and full Moon views. Use an adult-held torch and ball at three positions, keeping the light direction fixed and the ball slightly above head-shadow height. Compare the visible lit portion with the wheel.
- 2 min: Pens down if you are writing; pause and check. Ask an adult or chosen partner to check one view from the same observer position. Did the ball change shape, or did the visible lit part change?

- 6 min: Correct one phase label or view, explain what the model showed, and reset the room lighting and kit. A still-diagram explanation is an equal route.
- Access and equipment: Adult-prepared card circles, crayons, a ball and a low-power torch. Never point a torch into eyes; no looking at the Sun, lasers or blackout. Keep ordinary room light or use the supplied still model if dimming is unwelcome. An adult can operate all equipment.
- Drawing route: draw three equal circles; shade the part not seen lit for new, half and full. The circle itself stays round.
- Times to distinguish: one orbit relative to the stars takes about 27.3 days; the new-to-new phase cycle takes about 29.5 days. Both are roughly a month.
- NASA: Moon phases — <https://science.nasa.gov/moon/moon-phases/>
- NASA Space Place: Moon phases and orbital periods — <https://spaceplace.nasa.gov/moon-phases/>

Lundy Loop

- The torch model needs low light. You will get a warning before the lights change, every time.

Exit Ticket

- Does the Moon make its own light? Yes / No
- How long does the Moon take to orbit the Earth?
- About 27.3 days for one orbit, roughly a month. The phase cycle is about 29.5 days.
- Explain why phases are not caused by the Earth's shadow.
- Phases happen every month and depend on viewing angle; Earth's shadow only reaches the Moon during a rare eclipse.

Method and safe participation

1. Select the familiar evidence cards for the four science decisions.
2. Identify which observation belongs to each decision. Stop and check the task and agreed support with the adult.
3. Keep a first answer and use only the permitted support, with no new practical required.
4. In your book, record the evidence, a scientific conclusion and one checked improvement.

Only the apparatus for the current stage is on the table. If a pupil cannot safely take part, use a teacher demonstration and the same observation-and-conclusion task in the book.